PRACTICAL TASK FIVE

# DATABASE & FILESERVER CONNECTIVITY – ADD A NEW RECORD AND VERIFY PK

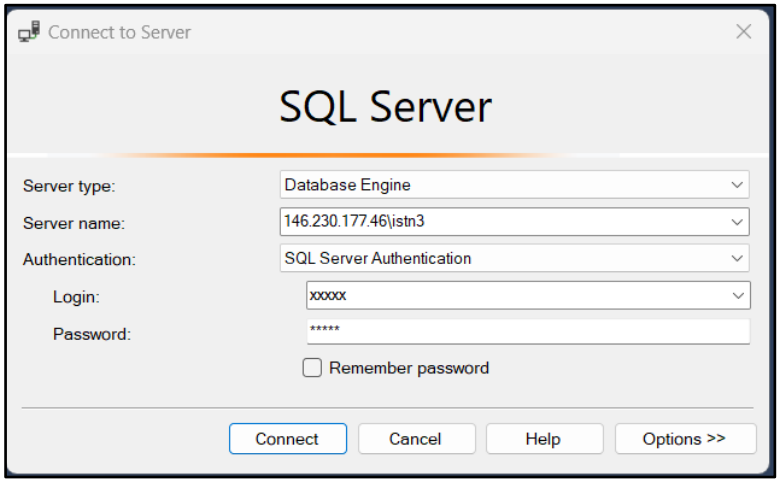
## OBJECTIVES

- Establish familiarity with Sql Server development tools
- Test out Sql Server individual accounts on Sql Server
- Create a simple database structures
- Populate database tables with data
- Create a C# application using Visual Studio (VS)
- Establish database connectivity with Sql Server database
- Display data from Sql Server using Windows Forms
- Add a new record and verify the PK of the newly inserted record

## PART A

1. Launch Sql Server Management Studio
2. Activate GlobalProtect (University VPN)
3. Establish a connection to the Sql Server
4. Note: The IP address for the SQL Server database is:

**146.230.177.46\ISTN3**

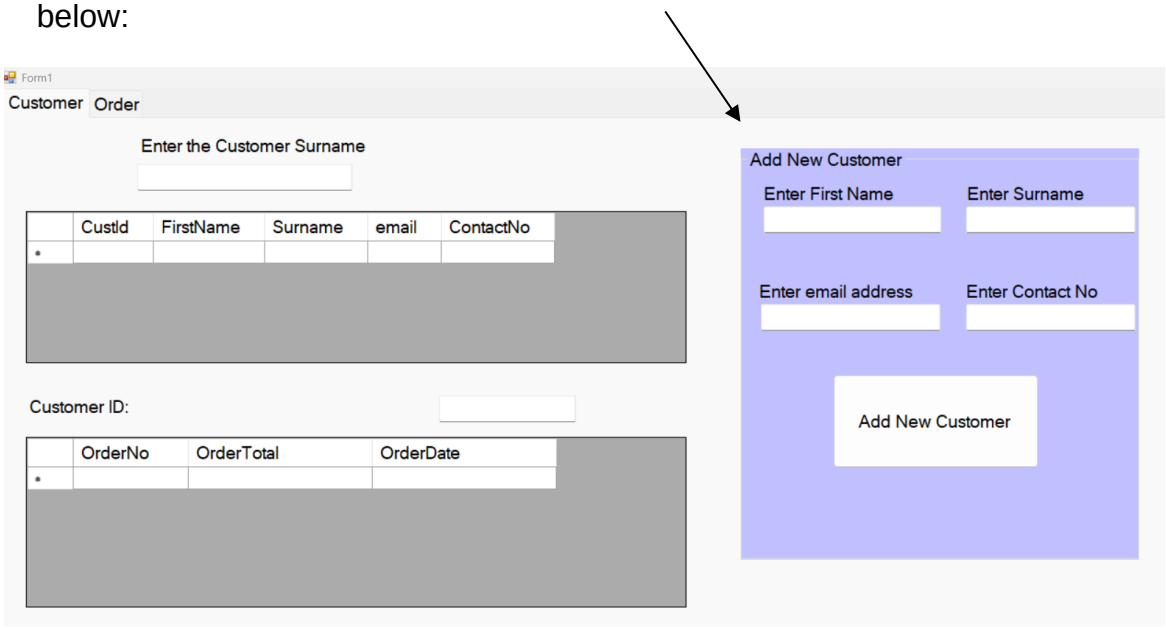


## PART B

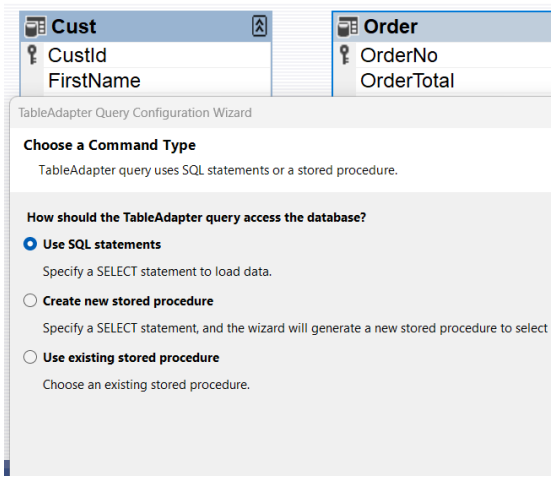
*This is a continuation of Prac 4*

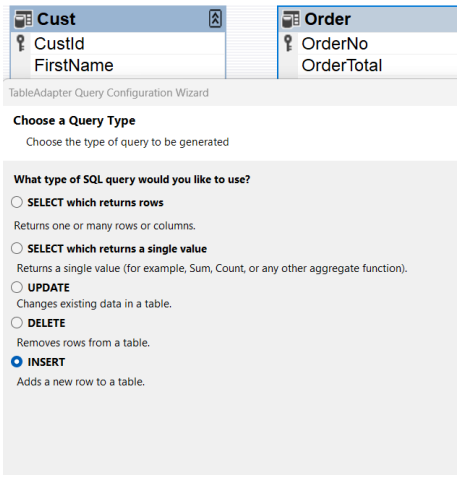
In Prac 4 you would have developed an app to display details of customers from a database table named Customer/Cust. Also, Prac 4 required the creation of a database table named Order. In the current prac both of these tables will be used to enable a Customer to record an order (overall order – not the order details). Prac 5 is a simple addition to the interface where a panel is created with the intention of providing the user with an opportunity to ADD A NEW CUSTOMER.

1. Create the Interface for adding a new Customer. It should look like the illustration below:

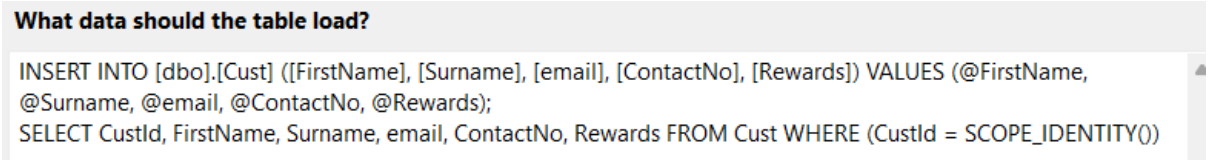



2. To Insert a New Customer, you can create a query on the table Customer table in the app's main dataset; the query is simply an Insert Sql query that takes in the customer details as parameters.



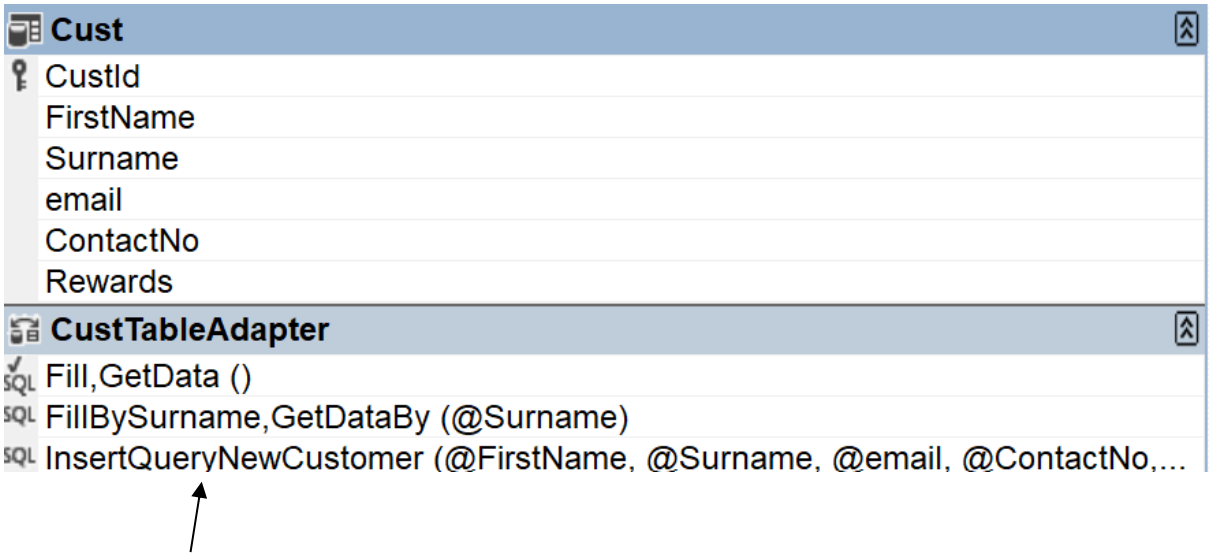



By default, a Sql Insert Query is automatically generated; it looks like…




This query is designed to Insert a New Customer into the database Customer table; he customer's Firstname, Surname, email, ContactNo and Rewards will be added as parameters when this sql query is implemented. The important thing to understand here is the 2nd part of the query which is optional (and can be deleted). However, the 2nd part makes use of the Scope_Identity() that is used to return to the application, the Primary Key of the newly inserted Customer record.

The Customer Table on the app's dataset now looks like…




A new InsertQuery has been added and it will function as a method that will insert a new customer. This method will insert a new customer into the database table and it will return the PK (ID) of the newly inserted customer

3. If you go back to the application code and add in the event handler for the button labelled "Add New Customer"

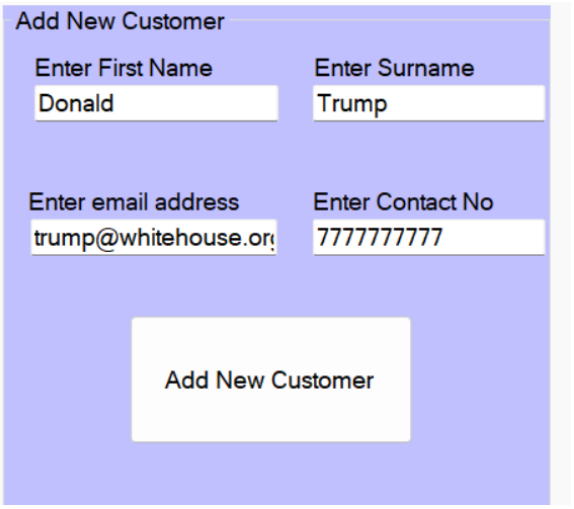
```
private void btnAddNewCust_Click(object sender, EventArgs e)
{
  int newCustID = (int)taCust.InsertQueryNewCustomer(txtFirst.Text, txtSurname.Text,
txtEmail.Text, txtContact.Text,null);
  taCust.Fill(dsCustOrder.Cust);
  MessageBox.Show("Customer " + txtFirst.Text+ " "+ txtSurname.Text+" has been added with
a Cust ID of: "+newCustID);
  txtFirst.Clear();
  txtSurname.Clear();
  txtEmail.Clear();
  txtContact.Clear();
}
```

This fragment of code inserts the new customer; returns the PK of the newly inserted customer; prints a confirmation message to indicate that the new customer has been inserted into the database table and it prints the ID of the new customer; once the insertion has taken place, all the fields are cleared so that the customer is not inserted into the database for a 2nd and third time. Once this code is working, you can add error handling routines where you check to ensure that all the customer details are provided before the insertion of the customer takes place. This can be done with if…then and try …catch statements and is left to you as an exercise. Also for the current app, the Rewards field for the Customer will be left blank – this is achieved by using the null value that is recognised in C#

4. One final part – if you would like to view the newly inserted customer inside the Customer database table, the you have to use the Customer table adapter and refill the customer gridview

```
taCust.Fill(dsCustOrder.Cust);
```

5. A sample run……





And if you take a look at the gridview….